

Richard (Kai) Hallett

Forward Deployed Engineer

Product engineering | applied AI | client delivery

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BASED / TRAVEL United Kingdom Remote, hybrid, willing to relocate	ENGINEERING 6.5 cumulative years Professional since April 2019	DELIVERY Discovery to production Support and handoff
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I make AI systems safe to trust where mistakes matter: fifteen years of clinical judgment under uncertainty, joined to production engineering. Product engineer with 6.5 cumulative years of professional software engineering since April 2019 across salaried roles, contracts, client delivery, and independent product development. I work best where the problem is still moving: embedding with the people affected, finding the operational constraint behind the brief, and carrying a useful system from discovery through production and handoff.

My recent focus has been applied AI and agentic systems, including evaluation, guardrails, fail-closed design, and provider-agnostic delivery. Before engineering, I worked as a cognitive behavioural therapist across NHS and private practice. That background transfers directly to high-stakes discovery, stakeholder trust, and noticing when confident output is wrong.

SELECTED ENGAGEMENTS

Loans by MAL - regulated UK lender, contract (2026). Turned regulatory constraints into a fail-closed conversational system: an untrusted LLM proposes each turn, a deterministic policy validator overrides unsafe output before it reaches a customer, and every turn is audit-traced. Hardened the engine against adversarial, confused, oversharing, and vulnerable personas before happy-path testing. Defined scope across concurrent epics, presented technical direction to C-suite stakeholders, and benchmarked providers on accuracy, cost, and safety behind a swappable adapter. Vue, TypeScript, Node.js, Zod, Terraform, AWS.

Swanage Traffic Alliance - community campaign (2025). Delivered a registration and campaign platform for a local traffic consultation, including an administration flow usable by non-technical volunteers. Astro, React, Vercel.

Becoming Diamond - coaching business (2025). Took the product from brief to revenue-ready: an animated marketing site and gated member portal with course content, AI chat, profiles, Stripe payments, and a git-based CMS the client could edit independently. Next.js, React 19, TypeScript, Tailwind, Stripe.

SELECTED SYSTEMS

LoanSlam - independent Phase 0 evaluation proof. A synthetic-scenario environment for the lending conversation engine, with 122 hostile scenarios, trace evidence, and a safety floor designed to report when the system is not ready. Vue, TypeScript, Node.js, Zod.

Sortie - ensemble AI code review. Three LLMs review a diff in parallel, a fourth synthesises, and merge is gated on convergent severity. Across six runs on a shipping codebase, 76.9% of distinct findings were surfaced by exactly one of the three reviewing models. One of those model-unique findings was a committed API key. TypeScript, Bun.

The Pit - agent evaluation and cost visibility. Structured multi-agent contests with observable traces, rubric scoring, failure tagging, and per-run microdollar cost accounting. Next.js, TypeScript, Postgres, Stripe.

■ EXPERIENCE

Oct 2024-present | Software Engineer | Oceanheart.ai, independent. Client-facing product delivery and independent engineering across applied AI, evaluation systems, commerce, content workflows, deployment, and support.

May-Jun 2026 | Full Stack Engineer | Loans by MAL, contract. Built the regulated conversational system above across Vue and TypeScript front end, Node.js and TypeScript backend, shared Zod contracts, deterministic safety validation, and Terraform on AWS.

Nov 2023-Mar 2024 | Software Engineer | EDITED, retail analytics. React, TypeScript, and Python data-visualisation features plus enterprise API integrations for an AI-driven retail analytics product.

Jun 2021-Nov 2023 | Software Engineer | Brandwatch, social intelligence. Built a greenfield enterprise data-visualisation platform in a team of five, modernised legacy Backbone code, and mentored junior developers.

Sep 2020-Feb 2021 | Full Stack Engineer | Telesoft Technologies, network security. TypeScript, Angular, Node.js, Express, and secure features for cybersecurity applications.

Mar-Aug 2020 | Web and Frontend Developer | Appius. CMS modules, customer portals, REST integrations, and responsive interfaces using .NET, JavaScript, and Vue.js.

Apr 2019-Mar 2020 | Software Developer | School Business Services. Vue.js front end for school financial-management software.

2004-2026 | Cognitive Behavioural Therapist | NHS and private practice. Worked across NHS and private practice; maintained a small private caseload alongside engineering.

■ EDUCATION AND TECHNICAL

EDUCATION

PGDip Cognitive Behavioural Therapy
- Royal Holloway, University of London

PGCert Primary Mental Healthcare -
University of Central Lancashire

BSc Psychology - UWE Bristol

TECHNICAL

Languages: TypeScript, Python, Go, Bash

AI engineering: agent and tool-layer architecture, evaluation harnesses, guardrails, fail-closed safety design, provider-agnostic multi-model pipelines, adversarial persona simulation, cost and token accounting

Product engineering: React, Next.js, Vue, Node.js, FastAPI, REST APIs, PostgreSQL, NATS

Cloud and delivery: AWS, Terraform, Docker, Kubernetes, Vercel, GitHub Actions, Grafana, Prometheus

Client-facing: discovery, scoping, C-suite communication, regulatory translation, production deployment, live support, operational handoff